



Lumen Build Tracker

Private desktop AI app · C:/Dev/tower-ai-app · Last updated Apr 20, 2026

⚡ **This session (Apr 20):** Agent routing verified, task results panel, auto-memory wiring, session log button, tool call UI overhaul, prompt caching, token truncation, Haiku for Helm, rolling context window, vault path UX, search confirmed live. 12 features shipped or confirmed.

31

Features shipped

5

In queue

v2

Current version

Tier 1

API tier (→ Tier 2 at \$40)

86% to daily-driver ready

SHIPPED — CORE INFRASTRUCTURE ✓ done



Electron + React + Tailwind v4 scaffold renderer

Multi-window Electron app, Vite bundler, Zustand stores, localStorage persistence, IPC bridge (preload.js → main.js)



Claude streaming API main

Full agent loop — text streaming, tool_use blocks, multi-turn tool results, abort support, error handling



Settings store + disk persistence store

All settings persisted via electron-store. API key, models, profile, theme, capabilities, vault path



Tool execution engine main

read_file, write_file, list_dir, run_command, browser_navigate, browser_get_content, browser_screenshot, browser_click, browser_type, web_search



Theme engine — 8 themes + dark/light/system renderer

Nord, Dracula, Gruvbox, Rose Pine, Monokai, Catppuccin, Tokyo Night, Lumen Dark. Per-theme CSS vars, density modes, font options



Chat + Code mode renderer

Conversation sidebar, message bubbles, markdown rendering, code blocks with copy/open-in-artifacts, streaming cursor, thinking blocks



Conversation search (Ctrl+K) renderer

Sidebar search bar with real-time filter. Collapses pinned/recents split when searching. Keyboard shortcut wired.



Helm mode — 4 agent routing renderer main

classifyPrompt routes to Code / Research / File / Schedule agent. Each gets a specialized system prompt. Dispatch indicator shows agent icon and live step name.



Scheduled tasks + cron runner main

Once / hourly / daily / weekly / monthly cadences. main.js cron engine runs tasks autonomously. Run-now via IPC. Results panel with history.



Memory system — inject + auto-generate store hook

Keyword-scored injection into system prompt. Auto-generation via Haiku after each turn (gated on toggle). Tags: fact / preference / project / person / context / note.



Skills system store

User-defined skills with keyword triggers. Auto-activate on message match. Injected into system prompt below memory block.



Projects — per-project system prompts store

Create projects, attach system prompts, pin conversations. Context injects into every message in that project.



Vault / Obsidian bridge main

vault:readFile, vault:writeFile, vault:listFiles, vault:search. Sandboxed to configured vault path. Session log writer uses this.



Session log button in chat header renderer

Notebook icon in conversation header. Calls Haiku to extract summary / decisions / next steps, writes to vault/Sessions/YYYY-MM-DD.md. Actionable vault-missing toast.



Tool call UI overhaul — grouped view renderer

All tool calls per message collapsed into one "✓ Searched N sources ▼" row. Expand to see clean list. Individual results on click. No more stacked boxes.



Token optimizations main hook

Prompt caching (anthropic-beta header + cache_control). Tool result truncation at 6k chars. Rolling 30-message context window. Haiku default for Helm.



Helm / Chat model split store

helmClaudeModel (default: Haiku, 50k TPM) separate from defaultClaudeModel (Sonnet, for chat). Configurable in Settings → Models.

IN QUEUE — NEXT UP planned



Obsidian folder indexing / RAG main

Auto-read and chunk vault files on startup. Surface relevant chunks in memory injection. Makes "use my Obsidian for context" work passively, not just on-demand.



Remote dispatch — phone → Lumen main

Lightweight HTTP listener in main.js. Accept prompts from Pushover / Ntfy / direct HTTP. Route through Helm agent system. Execute autonomously while away from desk.



Artifacts pane polish renderer

Side-by-side artifact view is functional but unstyled. Needs proper header, language label, copy button, full-screen toggle.



Token usage dashboard renderer

Settings → Usage shows monthly totals but no cache hit rate, no per-conversation breakdown, no cost estimate. Add cache hit % and estimated \$ cost.



System notifications for Helm tasks main

Scheduled tasks run in background but there's no OS notification when they complete. Wire Electron's Notification API to onCronTaskResult.

ARCHITECTURE NOTES reference

KEY FILES

`main.js` — Claude API, tools, cron, vault IPC
`preload.js` — window.tower bridge
`useClaudeStream.ts` — system prompt assembly, streaming
`helmStore.ts` — scheduled tasks, agents, results
`memoryStore.ts` — memory CRUD + generation + vault sync
`HelmPane.tsx` — all Helm UI (chat, dispatch, scheduled, agents)
`ToolCallCard.tsx` — grouped tool call display

SYSTEM PROMPT ORDER

1. **agentPrompt** — Helm agent specialization (most authoritative)
2. **profileBlock** — name, preferences, about
3. **projectPrompt** — active project system prompt
4. **memoryBlock** — top 8 keyword-matched memories
5. **skillsBlock** — triggered skill content

All cached via prompt-caching-2024-07-31 beta